MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys). My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.

This work aims to study, from an academical perspective at first, the subject of continuous measurement considering open quantum systems which was a different topic from my supervisor's specific area of research. Our primary motivation was therefore initially to educate ourselves on a field of quantum physics that is well known to a large part of the scientific community. By familiarizing ourselves with the topic, we decided to adopt a particular frame of report due to our journey of understanding. Therefore, the structure of this document is meant to help discovering and exploring the fundamentals of quantum continuous measurement to any unfamiliar reader. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts in the chapter concerning continuously monitored quantum open systems.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2]
However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

Chapter I

Basics of quantum mechanics

I.i Generalities on closed quantum systems

I.i.1 States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H}. \tag{I.1}$$

The Hamiltonian $\hat{H}$ is a specific Hermitian[1] operator representing classically the total energy of a system; this relationship will be detailed in Section I.i.2. Physically, the evolution of a system is governed by its energy, the *Schrödinger equation* states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle\,. \tag{I.2}$$

The solution of the Schrödinger equation can be expressed as an unitary[2] operator $\hat{U}$ such that,

$$|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle\,, \tag{I.3}$$

with initial condition $U(t_0,t_0) = \mathbb{I}$. By injecting the previous relation in the state equation, we get

$$\frac{\mathrm{d}\hat{U}}{\mathrm{d}t}(t,t_0) = -\frac{i}{\hbar}\hat{H}(t)\hat{U}(t,t_0). \tag{I.4}$$

We will simply assume the Hamiltonian to be time-independent in the following, nevertheless, solutions exist in the time-dependent case [3]. For us, the equation of evolution, solution of the Schrödinger equation, is

$$|\psi(t)\rangle = \underbrace{e^{-\frac{i\hat{H}(t-t_0)}{\hbar}}}_{\hat{U}(t,t_0)}\,|\psi(t_0)\rangle\,. \tag{I.5}$$

This solution is indeed **unitary** and we can use $\hat{U}^\dagger(t,t_0)$ on the left of both sides of the Eq. (I.5) to obtain $|\psi(t_0)\rangle = \hat{U}^\dagger(t,t_0)\,|\psi(t)\rangle$, i.e., we described **reversible** dynamics.

I.i.2 Observables

In quantum mechanics, any physical quantity is represented by an **observable** which is an operator fulfilling two properties: the eigenvectors of the observable form an orthonormal basis

[1] $\hat{H}^\dagger = \hat{H}$, where † denotes the conjugate transpose, defining the Hermitian adjoint.
[2] $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{I}$, where $\mathbb{I}$ is the identity.

of the Hilbert space $\mathcal{H}$, and, the corresponding eigenvalues are necessarily real [2]. In finite-dimensional Hilbert spaces, the definition of Hermitian operator[3] is sufficient for it to be an observable. In the case of infinite dimension Hilbert spaces, the definition of an observable requires more rigorous mathematical foundations (self-adjoint operators, the details of which lie beyond the purview of this study). In this sense, we only consider finite dimension cases in the following. We thus define the interval $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document, also, we lighten the notation by omitting $\hat{\bullet}$ symbols on operators hereafter.

A natural operator whose eigenbasis $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ is particularly significant is the Hamiltonian H, as its eigenvalues $\{E_n\}_{n\in I_{\mathcal{H}}}$ correspond to the allowed energy values of a system. We can understand the relation between an operator, its eigenvectors and eigenvalues with its corresponding eigenequation:

$$\forall n \in I_{\mathcal{H}}, \quad H\,|\phi_n\rangle = E_n\,|\phi_n\rangle\,. \tag{I.6}$$

Since $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ forms a basis of the Hilbert space, one can describe any state $|\psi\rangle$ in this very basis:

$$\forall\,|\psi\rangle \in \mathcal{H}, \qquad |\psi\rangle = \sum_{n\in I_{\mathcal{H}}} c_n\,|\phi_n\rangle\,, \tag{I.7}$$

where $c_n = \langle\phi_n|\psi\rangle$. This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian.

Finally, we mention the necessity of normalizing states, which is important when calculating probabilities (the Section I.ii makes it obvious). In order to normalize a state, one needs to set $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$, which is equivalent to the condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$.

I.i.3 Purity of states

The concept of *pure state* arises from the necessity to write the state of a system in a different way than that used so far in this document. Indeed, certain ways of preparing a system — such as a statistical mixture[4]— require another formalism because such processes prevent the state of the system from being expressed as a single $|\psi\rangle$ (or as a superposition of state vectors). To take into account such cases, we introduce the density matrix hereafter.

A density matrix ρ is an operator of the Hilbert space $\mathcal{H}$ (and also an operator of the dual space $\mathcal{H}^*$), which follows a set of properties:

- ρ is Hermitian: $\rho = \rho^\dagger$, implying it can be diagonalized and necessarily with a positive eigenvalues, $\mathrm{Sp}(\rho) \subset \mathbb{R}^+$.
- $\mathrm{Tr}\,[\rho] = 1$ necessarily.
- ρ is positive semi-definite: $\forall\,|\psi\rangle \in \mathcal{H}, \quad \langle\psi|\rho|\psi\rangle \geq 0$.

The density matrix formalism is particularly useful because it generalizes Dirac formalism. Indeed, if a state can be written as $|\psi\rangle$ then $\rho = |\psi\rangle\langle\psi|$ and we call it a **pure state**. Otherwise, the state is non-pure and can only be expressed with a density matrix. Also, pure states satisfy stronger properties than those defining a general density matrix; specifically, being idempotent $\rho^2 = \rho$ by construction, implying that $\mathrm{Tr}\left[\rho^2\right] = \mathrm{Tr}\,[\rho] = 1$ in the case of pure states.

[3]The eigenvectors of a Hermitian operator always span the entire Hilbert space (Spectral Theorem). Using the Gram-Schmidt process, we can always find a complete orthonormal basis from the operator's eigenspaces.

[4]In the Appendix A, we explain how to prepare a system as a statistical mixture or as a superposition of states. However, we recommend to read the Section I.ii on quantum measurement theory before.

I.i.4 Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathrm{End}(\mathcal{H})$ respecting properties I.i.3. For pure states we can describe them $|\psi\rangle \in \mathcal{H}$, or, by using the pure matrix density formalism $\rho_p = |\psi\rangle\langle\psi|$.
2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.
3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.
4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}[a_n]$ (which depends on the spectrum of the observable).
5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement collapses into the eigenstate $|\phi_n\rangle$.
6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the *Schrödinger* equation:
$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = -\frac{i}{\hbar}H(t)\,|\psi(t)\rangle\,. \tag{I.8}$$
For density matrices, the evolution is driven by the *Liouville-von Neumann* equation:
$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\left[H(t), \rho(t)\right]^{1}. \tag{I.9}$$

I.i.5 Multipartite systems

In quantum mechanics a system can be composed of multiple subsystems, we then call the global composition a multipartite system. The density matrix introduced in the previous sections is a useful tool to describe the state of such system and subsystems. If we consider an Hilbert space $\mathcal{H}_{\mathcal{AB}}$ composed of two physical systems A and B: $\mathcal{H}_{\mathcal{AB}} = \mathcal{H}_A \otimes \mathcal{H}_B$. Note that this implies $\dim(\mathcal{H}_{AB}) = \dim(\mathcal{H}_A) \times \dim(\mathcal{H}_B)$. The total system state is therefore represented by a density matrix $\rho_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$. The expression of the density matrix representing the state of the subsystem A is defined by
$$\rho_A = \mathrm{Tr}_B[\rho_{AB}], \tag{I.10}$$
and is called a *reduced density matrix*; moreover, we defined in this equation a new function called a *partial trace*. The idea behind the notation is to get rid of the B components in the tensor product of the total state ρ_{AB}. Explicitly, we can express the mean value of an operator of the subsystem A in the total space. Let $\hat{O} = \hat{O}_A \otimes \mathbb{I}_B$, and specify the bases of A and B: namely $\{|\phi_a\rangle\}_a$ and $\{|\phi_b\rangle\}_b$. The average of the operator $\hat{O}$ is expressed as
$$\begin{aligned}\left\langle \hat{O} \right\rangle = \mathrm{Tr}\left[\rho_{AB}\hat{O}\right] &= \sum_{a=1}^{\mathcal{H}_A}\sum_{b=1}^{\mathcal{H}_B} \langle\phi_a|\,\langle\phi_b|\,\rho_{AB}\left(\hat{O}_A \otimes \mathbb{I}_B\right)|\phi_b\rangle\,|\phi_a\rangle \\ &= \sum_{a=1}^{\mathcal{H}_A} \langle\phi_a| \underbrace{\sum_{b=1}^{\mathcal{H}_B} \langle\phi_b|\,\rho_{AB}\,|\phi_b\rangle}_{\rho_A} \hat{O}_A\,|\phi_a\rangle = \mathrm{Tr}_A\left[\rho_A\hat{O}_A\right].\end{aligned} \tag{I.11}$$

[1] $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is called the commutator of $\hat{A}$ and $\hat{B}$.

Therefore, for any local observable of A, the expectation value is encoded by the reduced density matrix ρ_A. Note that this average of a local observable does not require explicit knowledge on the total state, while the average still takes into account the correlations between A and B (due to $\rho_A = \mathrm{Tr}_B[\rho_{AB}]$).

The bipartite results shown above can be generalized to multipartite systems; however in the following, we will consider a system S coupled to an environnement E only, which is thus bipartite.

I.ii Quantum Measurement Theory (QMT)

In the QMT [6, 7], measuring consists in applying a projector to a state, so, we call this process a *projective measurement*. The set of projective measurements is called a Projection-Valued Measure (PVM) [8]. For the sake of simplicity, we work — as until now — with non-degenerate spectra for the observables. Since an observable has a set of eigenvectors $\{|a_n\rangle\}_{n\in I_{\mathcal{H}}}$, we can build a projector corresponding to one of them:

$$P_{a_n} = |a_n\rangle\langle a_n| \,. \tag{I.12}$$

This definition implies four important properties:

$$P_{a_n}^\dagger = P_{a_n}, \qquad P_{a_n}^2 = P_{a_n}, \qquad P_{a_n}P_{a_m} = P_{a_n}\delta_{nm}, \qquad \sum_n P_{a_n} = \mathbb{I}.$$

To represent the quantum mechanics definition of a measurement, the projective measurement needs to fulfill some requirements. The first requirement is to generate the probability of observing the result of a measurement on a system. Thus, the probability to observe a_n is defined by

$$\mathbb{P}_{|\psi\rangle}[a_n] = \langle\psi|P_{a_n}|\psi\rangle = |\langle\psi|a_n\rangle|^2, \tag{I.13}$$

$$\mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho P_{a_n}\right], \tag{I.14}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

$$\overline{|\psi_{a_n}\rangle} = P_{a_n}\,|\psi\rangle\,, \tag{I.15}$$

$$\overline{\rho_{a_n}} = P_{a_n}\rho P_{a_n}. \tag{I.16}$$

Here the $\overline{\bullet}$ symbol means that the state is not normalized, which will be an important notation for later, see Section II.iii. By normalizing we find the proper expression of a state after a measurement with result a_n:

$$|\psi_{a_n}\rangle = \frac{P_{a_n}\,|\psi\rangle}{\sqrt{\langle\psi|P_{a_n}|\psi\rangle}}, \tag{I.17}$$

$$\rho_{a_n} = \frac{P_{a_n}\rho P_{a_n}}{\mathrm{Tr}\left[\rho P_{a_n}\right]}. \tag{I.18}$$

The obtained state is said **conditioned by the measurement** result a_n, hence the importance of the index a_n on states obtained after measurement. We point out that the normalization factor is exactly the probability to observe the result a_n for the density matrix (or the square root of the probability, when using ket formalism). In the end, projective measurement fulfill the properties of quantum mechanics by construction because they generate probabilities (respecting the fact they need to be positive and their sum is 1), and can also be used to write the expression of collapsed states after measurement.

However, the definition of PVM has been found too restrictive to represent every kind of measurement allowed by quantum mechanics [9, 10]. One can define a more general kind of measurement, that fulfills quantum mechanics needs while relaxing some properties: the orthogonality between each measure elements is not required to generate probabilities. Therefore, one can build the so-called Positive Operator-Valued Measure (POVM), implicitly defined by the probability to observe the result a_n:

$$\mathbb{P}_\psi[a_n] = \langle\psi|\Pi_{a_n}|\psi\rangle\,, \tag{I.19}$$

$$\mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho\Pi_{a_n}\right], \tag{I.20}$$

where the operator Π_{a_n} is called the *effect* [7] for the result a. Moreover, the set of all effects $\{\Pi_{a_n}\}_{n\in I_\mathcal{H}}$ forms the POVM. As stated, the only properties we force the POVM to respect are:

$$\text{positive semi-definite: } \forall\,|\phi\rangle \in \mathcal{H},\ \langle\phi|\Pi_{a_n}|\phi\rangle \geq 0,\ \text{and,} \qquad \sum_{n\in I_\mathcal{H}} \Pi_{a_n} = \mathbb{I}.$$

These properties imply that POVM generate proper probabilities. Furthermore, the (normalized) states obtained after finding the result a_n are now,

$$|\psi_{a_n}\rangle = \frac{K_{a_n}|\psi\rangle}{\sqrt{\langle\psi|\Pi_{a_n}|\psi\rangle}}, \tag{I.21}$$

$$\rho_{a_n} = \frac{K_{a_n}\rho K_{a_n}^\dagger}{\mathrm{Tr}\left[\rho\Pi_{a_n}\right]}, \tag{I.22}$$

where we introduced the operator K_{a_n} called a *Kraus operator*; they respect the relation $\Pi_{a_n} = K_{a_n}^\dagger K_{a_n}$.

Also, POVM, just like PVM, allow one to describe the process of quantum measurement while *ignoring the results*. As in the thought experiment for statistical mixture of states in the Appendix A, we can describe statistically the state without knowing the measurement results:

$$\rho = \sum_{n\in I_\mathcal{H}} \mathbb{P}[a_n]\rho_{a_n} = \sum_{n\in I_\mathcal{H}} K_{a_n}\rho K_{a_n}^\dagger. \tag{I.23}$$

The corresponding relation for PVM is obtained by replacing both K_{a_n} and $K_{a_n}^\dagger$ by P_{a_n}.

Finally, one can observe that POVM generalize PVM in the sense that projectors are special cases of POVM elements (adding self-adjoint property and orthogonality to POVM gives a PVM). Furthermore, PVMs make states collapse on eigenstates that are necessarily orthogonal to each other, which is not necessarily the case for POVMs. A brief example of POVM forcing a state to collapse onto "non-orthogonal states" is when considering initially the state $|\psi\rangle = \frac{|g\rangle+|e\rangle}{\sqrt{2}} \equiv |+\rangle$. If we choose the Kraus operators $K_0 = |g\rangle\langle g|$ and $K_1 = |+\rangle\langle e|$, which fulfill $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$, and, that form the POVM $\{|g\rangle\langle g|\,, |e\rangle\langle e|\}$. Then, when the result of the measurement is 0, we find that the state collapses to $|\psi_0\rangle = |g\rangle$, and when finding 1 it collapses to $|\psi_1\rangle = |+\rangle$. However, $\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}} \neq 0$, thus the states on which the initial state can collapse are not orthogonal to each other.

I.iii Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. However, this description applies to the ideal case where the quantum system is considered isolated from any external interaction. Instead, in the OQS theory systems interact

locally with an environment, which itself interacts with a wider environment, etc... To avoid considering infinite chains of interaction, the theory truncates the environment to a local vicinity of the system. Thus, we approximate the total system T as a closed quantum system, within which the quantum system of interest S is called an open quantum system. Moreover, as explained S couples with its environment E, thus giving rise to a dynamical regime distinct from that of closed quantum systems.

I.iii.1 Building the formalism

We will present here a derivation of the Lindblad master equation using the *input-output formalism* [5, 11]. This derivation can alternatively be carried out using a microscopic approach [3]. Since a full derivation of the input-output formalism is mathematically challenging, we will only provide its key equations and their physical meaning. Further details on the input-output theory and quantum stochastic calculus are available in Appendix C.

As seen in Section I.i.5, the Hilbert space of the total system is divided into $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. We consider as initial state a factorized state: $\rho_T = \rho_S \otimes \rho_E$. From now on we will use a typographic convention where $\rho_T \equiv \boldsymbol{\varrho}$ and $\rho_S \equiv \rho$, such that the last assumption rewrites $\boldsymbol{\varrho} = \rho \otimes \rho_E$. The system Hamiltonian H_S depends on which physical system is studied and is left free. Instead, the environment is chosen as a collection of quantum harmonic oscillators, the environment Hamiltonian being thus defined by

$$H_E = \int_0^{+\infty} \hbar\omega b_\omega^\dagger b_\omega \mathrm{d}\omega, \tag{I.24}$$

where the operators of the environment satisfy $\left[b_\omega, b_{\omega'}^\dagger\right] = \delta(\omega - \omega')$. This Hamiltonian represents a continuum of bosonic modes of frequency $\omega \in \mathbb{R}^+$, where each mode is represented by a quantum harmonic oscillator. We define the so-called interaction Hamiltonian by

$$H_I = i\hbar \int_0^{+\infty} \sqrt{\frac{k(\omega)}{2\pi}} \left(c b_\omega^\dagger - c^\dagger b_\omega\right) \mathrm{d}\omega, \tag{I.25}$$

where c is an operator of the system, and $\omega \mapsto k(\omega)$ is a function representing the coupling strength between S and E for each considered mode at frequency ω. This coupling is treated as a perturbation: the interaction has a smaller impact on the dynamics than the free evolution. This form of interaction Hamiltonian can be found, for example, in the case of a two-level system interacting with a radiation field via a dipole interaction, once the rotating wave approximation has been performed. Consequently by using Eq. (I.24,I.25), we define the total Hamiltonian:

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{I.26}$$

By switching to the interaction picture (detailed in Appendix B), we can get rid of the free dynamics governed by H_0. Indeed, in the interaction picture the Liouville-von Neumann equation becomes

$$\frac{\mathrm{d}\tilde{\boldsymbol{\varrho}}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I, \tilde{\boldsymbol{\varrho}}\right], \tag{I.27}$$

in which we used the following notation: for any operator $O(t)$ we denote by $\tilde{O}(t)$ its expression in the interaction picture, such that $\tilde{O}(t) = U_0^\dagger(t, t_0) O(t) U_0(t, t_0)$. The unitary operator U_0 is the free evolution operator generated by H_0, and is defined as $U_0(t, t_0) = e^{-\frac{i}{\hbar} H_0 (t - t_0)}$.

Before calculating explicitly the expression of $\tilde{H}_I$, we need to make some assumptions which are listed below.

- The *first Markovian assumption* we consider the interaction strength to be constant in a (large) bandwidth $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$ (ω_0 is the natural frequency of S), so that $\forall \omega \in \mathcal{W}, \quad k(\omega) = k$. The coupling strength is considered negligible outside of $\mathcal{W}$.
- The system operator acquires a time-dependent phase in the interaction picture: $\tilde{c}(t) = ce^{-i\omega_0(t-t_0)}$. This ansatz is motivated by the explicit form of operators in the interaction picture when considering elementary systems [5].
- We assume that $k \ll \omega_0$, which establishes $\tau = \frac{1}{k}$ as the dominant timescale for the dynamics.

We can now express the interaction Hamiltonian in the interaction picture by

$$\tilde{H}_I(t) = i\hbar\sqrt{k}\Big[cb_t^\dagger - c^\dagger b_t\Big], \tag{I.28}$$

in which we defined the input field operator b_t as

$$b_t = \frac{1}{\sqrt{2\pi}} \int_{\mathcal{W}} b_\omega e^{-i(\omega-\omega_0)(t-t_0)} \mathrm{d}\omega. \tag{I.29}$$

To characterize the environmental operators b_t, we can compute their canonical commutation relation:

$$\Big[b_t, b_{t'}^\dagger\Big] = \frac{1}{\sqrt{2\pi}} \int_{-\theta}^{\theta} e^{-i\omega(t-t')} \mathrm{d}\omega = \delta(t-t'). \tag{I.30}$$

To obtain Eq. (I.30), we assumed the frequency bandwidth $\mathcal{W}$ to be much larger than all other frequencies of the dynamics, implying in particular $k \ll \theta$. In this limit, the integration can be extended to the entire real line, which yields the definition of the Dirac delta distribution. Note that by defining $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$, we implicitly state that $\omega_0 \geq \theta$. This condition gives again the third assumption mentioned earlier: $k \ll \theta \leq \omega_0 \Rightarrow k \ll \omega_0$. The delta-distribution we just obtained in Eq. (I.30) is not satisfying because the commutator becomes singular at equal times. We will now open a brief parenthesis on input operators and Wiener increments to bypass this singularity.

We define a Wiener process in quantum stochastic calculus as

$$W(t, t_0) = \int_{t_0}^{t} b_{t'} \mathrm{d}t'. \tag{I.31}$$

We can further derive a differential quantity using calculus [12], called a quantum Wiener increment and defined as

$$\mathrm{d}W_t = \lim_{\delta t \to \mathrm{d}t} [W(t + \delta t, t_0) - W(t, t_0)] = b_t \mathrm{d}t. \tag{I.32}$$

Using Eq. (I.31), the quantum Wiener increment canonical commutation relation (at same times) is

$$\Big[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\Big] = \mathrm{d}t. \tag{I.33}$$

However, by calculating directly the same commutator with the right hand side of Eq. (I.32), we find

$$\Big[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\Big] = \Big[b_t, b_t^\dagger\Big] \mathrm{d}t^2. \tag{I.34}$$

When we combine this result with Eq. (I.33), we get

$$\Big[b_t, b_t^\dagger\Big] \mathrm{d}t = \mathbb{I}. \tag{I.35}$$

If we now define a set of bosonic operators B_t such that $B_t = b_t\sqrt{\mathrm{d}t}$, then

$$\left[B_t, B_t^\dagger\right] = \mathbb{I}, \tag{I.36}$$

which is the canonical commutation relation of creation/annihilation bosonic operators[5]. Since $\{B_t\}_{t\in\mathbb{R}}$ forms an infinite set of proper annihilation operators, the environment can be modeled as a continuous collection of quantum harmonic oscillators. In this framework, at any time t, the system couples to the corresponding temporal bosonic mode defined by B_t. After evolving over the infinitesimal duration $\mathrm{d}t$, the system decouples from the mode at t and interacts with the next mode at $t + \mathrm{d}t$.

We now make use of one more assumption to derive the evolution of the state $\rho(t)$ of the open quantum system S.

- The *Born-Markov approximation*: at each time t, the total state $\boldsymbol{\varrho}(t)$ is a factorized state with the "same" reduced state for the environment, explicitly

$$\forall t \in \mathbb{R}, \quad \tilde{\boldsymbol{\varrho}}(t) = \tilde{\rho}(t) \otimes \nu_t, \tag{I.37}$$

  where $\nu_t = |0\rangle\langle 0|_t$ is the vacuum state of the bosonic operator B_t. Coupling with the corresponding vacuum state (at time t) implies that the average of the input modes canonical anticommutator[6] is

$$\left\langle \left\{ b_t, b_{t'}^\dagger \right\} \right\rangle = \delta(t - t'). \tag{I.38}$$

  This equation and Eq. (I.30) form the Markovian assumptions of this framework [5]: the bosonic mode B_t is completely uncorrelated with the other modes at different times due to the delta-correlation of b_t. The global state evolves over an infinitesimal time $\mathrm{d}t$ as

$$\tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t) = U_I(t + \mathrm{d}t, t) \overbrace{\left[\tilde{\rho}(t) \otimes \nu_t\right]}^{\tilde{\boldsymbol{\varrho}}(t)} U_I^\dagger(t + \mathrm{d}t, t), \tag{I.39}$$

  where $U_I(t + \mathrm{d}t, t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)dt}$, as explained in Appendix B.

I.iii.2 The Lindblad master equation

We just calculated the evolution of the state $\boldsymbol{\varrho}$ of the total system over an infinitesimal time $\mathrm{d}t$. In order to obtain the evolved state $\rho(t + \mathrm{d}t)$ of the system S, we want to "trace out" the environment (as explained in Section I.i.5) by applying the following partial trace

$$\tilde{\rho}(t + dt) = \mathrm{Tr}_E[\tilde{\boldsymbol{\varrho}}(t + \mathrm{d}t)]. \tag{I.40}$$

To write this expression into explicit terms, we develop the evolved total state and thus, we need to explicit the expression of $U_I(t + \mathrm{d}t, t)$. To do so, we can rewrite it using the environmental ladder operators, using Eq. (I.28), as

$$U_I(t + \mathrm{d}t, t) = \exp\left[\sqrt{k\mathrm{d}t}\Big(c \otimes B_t^\dagger - c^\dagger \otimes B_t\Big)\right]. \tag{I.41}$$

The subtlety here is that the unitary evolution operator scales with $\sqrt{\mathrm{d}t}$, and because $\mathrm{d}t = o(\sqrt{dt})$ expanding to order $\mathrm{d}t$ is already a fine description of the dynamics. Consequently, we expand

[5] In the input-output formalism, the annihilation operator acts on its Fock basis as $B_t\,|n\rangle_t = \sqrt{n}\,|n-1\rangle_t$ (with the vacuum condition $B_t\,|0\rangle_t = 0$), whereas the adjoint (creation operator) acts as $B_t\,|n\rangle_t = \sqrt{n+1}\,|n+1\rangle_t$.

[6] We define the anticommutator between two operators A and B as $\{A, B\} = AB + BA$.

the exponential to second order so that we capture all terms up to $\mathrm{d}t$, giving

$$U_I(t+\mathrm{d}t,t) \approx \mathbb{I}\otimes\mathbb{I} + \Big(c\otimes B_t^\dagger - c^\dagger\otimes B_t\Big)\sqrt{k\mathrm{d}t} + \Big(c^{\dagger 2}\otimes B_t^2 + c^2\otimes B_t^{\dagger 2} - cc^\dagger\otimes B_t^\dagger B_t - c^\dagger c\otimes B_t B_t^\dagger\Big)\frac{k\mathrm{d}t}{2}. \quad \text{(I.42)}$$

We can now rewrite the evolution of $\tilde{\boldsymbol{\varrho}}$ during the infinitesimal time $\mathrm{d}t$, i.e. Eq. (I.39), using the previous expansion:

$$\tilde{\boldsymbol{\varrho}}(t+\mathrm{d}t) \approx [\tilde{\rho}(t)\otimes|0\rangle\langle 0|_t]\times 1 + \Big[\tilde{\rho}(t)c^\dagger\otimes|0\rangle\langle 1|_t + \text{h.c.}\Big]\times\sqrt{k\mathrm{d}t} + \left[\frac{\sqrt{2}}{2}\tilde{\rho}(t)c^{\dagger 2}\otimes|0\rangle\langle 2|_t + \text{h.c.} - \frac{1}{2}\tilde{\rho}(t)cc^\dagger\otimes|0\rangle\langle 0|_t + \text{h.c.} + c\tilde{\rho}(t)c^\dagger\otimes|1\rangle\langle 1|_t\right]\times k\mathrm{d}t, \quad \text{(I.43)}$$

where "h.c." stands for hermitian conjugate of the last term. We can now apply the partial trace, the evolved state of the system S is thus expressed as

$$\tilde{\rho}(t+\mathrm{d}t) = \tilde{\rho}(t) + k\left[c\tilde{\rho}(t)c^\dagger - \frac{c^\dagger c\tilde{\rho}(t) + \tilde{\rho}(t)c^\dagger c}{2}\right]\mathrm{d}t.$$

If we now form the difference quotient $\frac{\tilde{\rho}(t+\mathrm{d}t)-\tilde{\rho}(t)}{dt}$, in the infinitesimal limit, we find the differential equation followed by the state ρ at all time

$$\frac{\mathrm{d}\tilde{\rho}}{\mathrm{d}t}(t) = k\mathfrak{D}[c]\tilde{\rho}(t), \quad \text{(I.44)}$$

in which we used the definition of the super-operator $\mathfrak{D}[\hat{O}]\bullet \equiv \hat{O}\bullet\hat{O}^\dagger - \frac{\{\hat{O}^\dagger\hat{O},\bullet\}}{2}$, called the dissipator. Finally, by coming back into the Schrödinger representation, we find the so-called Lindblad master equation:

$$\boxed{\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}[H_0,\rho(t)] + k\mathfrak{D}[c]\rho(t)}. \quad \text{(I.45)}$$

Chapter II

Continuously measured open quantum systems

In the previous chapters, we went through the steps leading to the Lindblad master equation using the input-output formalism. The environment has then been traced out to recover information on the system S from the global coupled system, leaving an impact on the dynamics of S crystallized in the expression of a dissipator $\mathfrak{D}$. In the following, we will continuously monitor the environment by measuring the environmental quantum oscillator. Moreover, two kind of measurements will be presented below: namely the photo-detection and the homodyne detection. In each case, the reader will be able to find graphical illustrations accompanying the physical equations, which have been generated in `Julia` using the packages `QuantumToolbox.jl` to simulate dynamics, and `Plots.jl` to generate graphs. In order to lighten notations, we stop specifying by $\tilde{\bullet}$ the interaction picture, however if a change of representation occurs, it will be specified. Furthermore, we work from now on in natural units where $\hbar = 1$.

We should now calculate the expression of the conditional state obtained at time $t + \mathrm{d}t$. Let $\{|j\rangle\langle j|_t\}_j$ be a PVM (see Section I.ii). The measurement projects the environment with corresponding oscillator B_t on the PVM element $|j\rangle\langle j|_t$. After the measurement on the total state and after applying the partial trace Tr_E, we find

$$\rho_j(t + \mathrm{d}t) = \frac{\langle j|_t \, \boldsymbol{\varrho}_c(t + \mathrm{d}t) \, |j\rangle_t}{p_j(t + \mathrm{d}t)}, \tag{II.1}$$

where $p_j(t + \mathrm{d}t) = \mathrm{Tr}\left[\boldsymbol{\varrho}_c(t + \mathrm{d}t)(\mathbb{I} \otimes |j\rangle\langle j|_t)\right]$. Note that in this expression

$$\boldsymbol{\varrho}_c(t + \mathrm{d}t) = U_I(t + \mathrm{d}t, t) \overbrace{\left[\rho_c(t) \otimes |0\rangle\langle 0|_t\right]}^{\boldsymbol{\varrho}_c(t)} U_I^\dagger(t + \mathrm{d}t, t), \tag{II.2}$$

where the subscript c indicates the system state is conditional. If the condition is obvious, we choose not to write this subscript explicitly, instead we write for example ρ_j. The system in this calculation is indeed conditional because we consider the measurement already present in the evolution before time t.

II.i Photodetection

II.i.1 Unravelling the Stochastic Master Equation (SME)

In the case where an open quantum system is continuously monitored by a photo-detector, the mathematical impact is a projection of the environment on the corresponding Fock basis

$\{|n\rangle\langle n|_t\}_{n\in\mathbb{N}}$: we actually measure the environment and not the system. The Eq. (I.iii.2) shows that only projections on $|0\rangle\langle 0|_t$ and $|1\rangle\langle 1|_t$ contribute to the result, which will be shown hereafter for each measurement result separately.

Measurement result 0:

From the expansion in Eq. (I.iii.2), we find

$$\rho_0(t+\mathrm{d}t) = \frac{\langle 0|_t\, \boldsymbol{\varrho}_c(t+\mathrm{d}t)\, |0\rangle_t}{p_0(t+\mathrm{d}t)} = \frac{\rho_c(t) - \left\{\rho_c(t), c^\dagger c\right\}\frac{k\mathrm{d}t}{2}}{p_0(t+\mathrm{d}t)}. \tag{II.3}$$

The normalization is found using the same expansion, giving

$$p_0(t+\mathrm{d}t) = 1 - \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t, \tag{II.4}$$

where we defined the average labelled by t as $\left\langle \hat{O}\right\rangle_t = \mathrm{Tr}\left[\rho_c(t)\hat{O}\right]$. In the infinitesimal limit, we can also expand the denominator in the evolved conditional state, giving

$$\rho_0(t+\mathrm{d}t) \approx \bar{\rho}_0(t+\mathrm{d}t) \times \left(1 - \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t\right), \tag{II.5}$$

note that we used the same notation as in Section I.ii, where $\bar{\rho}$ indicates an unnormalized state (here in the interaction picture in particular). Finally, by introducing the measurement super-operator $\mathfrak{H}[\hat{O}]\bullet = \hat{O}\bullet + \bullet\hat{O}^\dagger - \left\langle \hat{O} + \hat{O}^\dagger\right\rangle_t \bullet$, we can write that

$$\rho_0(t+\mathrm{d}t) = \rho_c(t) - \frac{k\mathrm{d}t}{2}\mathfrak{H}\left[c^\dagger c\right]\rho_c(t). \tag{II.6}$$

Measurement result 1:

The same calculation method applies and the conditional (unnormalized) state is expressed as

$$\bar{\rho}_1(t) = \langle 1|_t\, \boldsymbol{\varrho}(t+dt)\, |1\rangle_t = c\rho_c(t)c^\dagger k\mathrm{d}t. \tag{II.7}$$

The normalization is then

$$p_1(t+\mathrm{d}t) = \left\langle c^\dagger c\right\rangle_t k\mathrm{d}t. \tag{II.8}$$

Combined, they give the normalized conditional state

$$\rho_1(t+\mathrm{d}t) = \frac{c\rho_c(t)c^\dagger}{\langle c^\dagger c\rangle_t}. \tag{II.9}$$

Mixing the results:

If we observe from a statistical point of view the last results, we have a measurement outcome at time $t+\mathrm{d}t$ that can be 0 with a probability $p_0(t+\mathrm{d}t)$ or 1 with probability $p_1(t+\mathrm{d}t)$. We can imagine a process where we count how many times the result 1 appears, storing the result in a random variable $N(t,t_0)$. In an infinitesimal slice of time $\mathrm{d}t$, the increment $\mathrm{d}N_t = N(t+\mathrm{d}t,t_0) - N(t,t_0)$ is either 1 or 0, and so we have the fundamental relation $\mathrm{d}N_t^2 = \mathrm{d}N_t$.

These are respectively called Poisson process and Poisson increment[1]. For now, we can calculate the stochastic average of $\mathrm{d}N_t$ to get

$$\mathbb{E}[\mathrm{d}N_t] = 0 \times p_0(t+\mathrm{d}t) + 1 \times p_1(t+\mathrm{d}t) = \left\langle c^\dagger c \right\rangle_t k\mathrm{d}t. \tag{II.10}$$

Note that by experimentally averaging $\mathrm{d}N_t$, we find the expectation value $\left\langle c^\dagger c \right\rangle_t$ which is the mean occupation number at time t of the system S. This Poisson increment will allow us to write the differential conditional state $\mathrm{d}\rho_c(t+\mathrm{d}t)$ with respect to each possible result, because $\mathrm{d}N_t = 0$ means the differential is $\mathrm{d}\rho_0(t) = \rho_0(t+\mathrm{d}t) - \rho_c(t)$, instead $\mathrm{d}N_t = 1$ means we the differential is $\mathrm{d}\rho_1(t) = \rho_1(t+\mathrm{d}t) - \rho_c(t)$ state. Indeed,

$$\mathrm{d}\rho_c(t) = \mathrm{d}N_t[\rho_1(t+\mathrm{d}t) - \rho_c(t)] + (1-\mathrm{d}N_t)[\rho_0(t+\mathrm{d}t) - \rho_c(t)]. \tag{II.11}$$

We can inject in this last equation the results of Eq. (II.6,II.9) to get (in the Schrödinger picture),

$$\boxed{\mathrm{d}\rho_c = -i[H_0,\rho_c]\mathrm{d}t - \frac{k}{2}\mathfrak{H}\left[c^\dagger c\right]\rho_c\mathrm{d}t + \left(\frac{c\rho_c c^\dagger}{\langle c^\dagger c\rangle_t} - \rho_c\right)\mathrm{d}N_t}, \tag{II.12}$$

in which we discarded terms of order $\mathrm{d}N_t\mathrm{d}t$ (this can be understood informally because $\mathbb{E}[\mathrm{d}N_t] = \lambda_t\mathrm{d}t$ and $\mathbb{V}[\mathrm{d}N_t] \approx \lambda_t\mathrm{d}t$). This equation is called the photo-detection Stochastic Master Equation (SME). The importance of this equation is double because it contains in average the **unconditional** dynamics: $\rho(t) = \mathbb{E}[\rho_c(t)]$. Finally, when derived for a pure state the expression simplifies to a Stochastic Schrödinger Equation (SSE):

$$\mathrm{d}\left|\psi_c\right\rangle = \left[-iH_0\mathrm{d}t + \frac{k}{2}\left(\left\langle c^\dagger c\right\rangle_t - c^\dagger c\right)\mathrm{d}t + \left(\frac{c}{\sqrt{\langle c^\dagger c\rangle_t}} - 1\right)\mathrm{d}N_t\right]\left|\psi_c\right\rangle, \tag{II.13}$$

which conserves the purity of the state during the stochastic evolution. Note that we can derive the SME from the SSE using Itō calculus rule

$$\mathrm{d}\rho_c = \mathrm{d}\left|\psi_c\right\rangle\left\langle\psi_c\right| + \left|\psi_c\right\rangle\mathrm{d}\left\langle\psi_c\right| + \mathrm{d}\left|\psi_c\right\rangle\mathrm{d}\left\langle\psi_c\right|. \tag{II.14}$$

II.i.2 The unconditional evolution

Physically, averaging over all possible conditional trajectories gives the unconditional trajectory (as if no measurement was ever done). We can show this by using the average in Eq. (II.10) to find

$$\mathbb{E}[\mathrm{d}\rho_c] = -i[H_0,\rho]\mathrm{d}t + k\mathfrak{D}[c]\rho\mathrm{d}t, \tag{II.15}$$

which is exactly the Lindblad master equation. We can see the conditional trajectory as one path followed by the system among the infinite possibilities. In this sense we will refer to ρ_c as an *unravelling of the source dynamics* (or a *quantum trajectory*) since it decomposes the solutions of the Lindblad master equation [13]. Therefore, it is common to call the process of finding the expression of a quantum trajectory as *an* unravelling of the master equation; here we presented an unravelling based on continuous jump evolution.

[1] The Poisson increment follows a Poisson distribution with rate $\Lambda_t \in \mathbb{R}$: $\forall n \in \mathbb{N}$, $\mathbb{P}[\mathrm{d}N_t = n] = \frac{\Lambda_t^n}{n!}e^{-\Lambda_t}$. However because we only have possible outcomes (0 or 1), the parameter is forced to be $\Lambda_t = 0$ in our case. We can state that $\mathbb{P}[\mathrm{d}N_t = 1] = \lambda_t\mathrm{d}t \approx 0$ and $\mathbb{P}[\mathrm{d}N_t = 0] = 1 - \lambda_t\mathrm{d}t \approx 1$ to obtain the Poisson distribution results on an infinitesimal scale. It is trivial that $\mathbb{E}[\mathrm{d}N_t] = \lambda_t\mathrm{d}t$, which corresponds to results found in Eq (II.10).

II.ii Homodyne detection

In quantum optics [14], the *quadrature operator* is well-known and a useful operator REWRITE that we use in this section to measure the environment continuously, when doing so we call the physical process a *homodyne measurement*. The quadrature operator is defined by $\hat{x}_\theta = (e^{-i\theta}B_t + e^{i\theta}B_t^\dagger)/\sqrt{2}$ where $\theta \in \mathbb{R}$, and verifies $\left[\hat{x}_\theta, \hat{x}_{\theta+\frac{\pi}{2}}\right] = i$. Even if it is not possible to construct normalized eigenstates of this operator, we can build the states $|x_\theta\rangle$ satisfying $\hat{x}_\theta |x_\theta\rangle = x_\theta |x_\theta\rangle$ such that $|x_\theta\rangle\langle x_\lambda| = \delta(\theta - \lambda)$, verifying $\int_{\mathbb{R}} |x_\theta\rangle\langle x_\theta| \, \mathrm{d}x_\theta = 1$. We will use the following overlapping property [14] between these states and the Fock states:

$$\langle x_\theta | n \rangle_t = \frac{e^{-\frac{x_\theta}{2}} H_n(x) e^{-in\theta}}{\sqrt{2^n n!} \pi^{\frac{1}{4}}}, \tag{II.16}$$

where $H_n(x)$ is the Hermite polynomial of order n. In particular we will later use the fact that

$$\langle x_\theta | 0 \rangle_t = \pi^{-\frac{1}{4}} e^{-\frac{x_\theta^2}{2}}, \text{ and that } \langle x_\theta | 1 \rangle_t = \sqrt{2} x_\theta e^{-i\theta} \langle x_\theta | 0 \rangle_t . \tag{II.17}$$

Note that because the state of the environnement at time t is the corresponding vacuum state, we have

$$p_{x_\theta}(t) = |\langle x_\theta | 0 \rangle_t|^2 = \frac{e^{-x_\theta^2}}{\sqrt{\pi}} \tag{II.18}$$

In order to express the continuous homodyne measurement, the environment is projected at each time on the quadrature states, giving the conditional (unnormalized) state

$$\begin{aligned} \bar{\rho}_c(t + \mathrm{d}t) &= \mathrm{Tr}\Big[U_I(t + \mathrm{d}t, t) \boldsymbol{\varrho}_c(t) U_I^\dagger(t + \mathrm{d}t, t)(\mathbb{I} \otimes |x_\theta\rangle\langle x_\theta|)\Big] \\ &= \langle x_\theta | U_I(t + \mathrm{d}t, t)[\rho_c(t) \otimes |0\rangle\langle 0|_t] U_I^\dagger(t + \mathrm{d}t, t) | x_\theta \rangle , \end{aligned} \tag{II.19}$$

with a corresponding normalization (probability)

$$p_{x_\theta}(t + \mathrm{d}t) = \mathrm{Tr}\left[\bar{\rho}_c(t + \mathrm{d}t)\right]. \tag{II.20}$$

We will now use the expansion of the Eq. (I.iii.2), however we truncate it to the most dominant terms ($\sqrt{dt}$), giving for the conditional unnormalized evolved state

$$\bar{\rho}_c(t+\mathrm{d}t) = |\langle x_\theta | 0 \rangle_t|^2 \rho_c(t) + \Big[c\rho_c(t) \langle x_\theta | 1 \rangle_t \langle 0 | x_\theta \rangle + \rho_c(t) c^\dagger \langle x_\theta | 0 \rangle_t \langle 1 | x_\theta \rangle\Big] \sqrt{k\mathrm{d}t} + o\Big(\sqrt{\mathrm{d}t}\Big). \tag{II.21}$$

Simplifying this expression and factorizing it using Eq. (II.17,II.18), we find

$$\bar{\rho}_c(t + \mathrm{d}t) = p_{x_\theta}(t)\Big\{\rho_c(t) + x_\theta \sqrt{2k\mathrm{d}t}\Big[c\rho_c(t)e^{-i\theta} + \rho_c(t)c^\dagger e^{i\theta}\Big]\Big\} + o\Big(\sqrt{dt}\Big), \tag{II.22}$$

with a normalization

$$p_{x_\theta}(t + \mathrm{d}t) = p_{x_\theta}(t)\Big\{1 + x_\theta \sqrt{2k\mathrm{d}t} \Big\langle ce^{-i\theta} + c^\dagger e^{i\theta}\Big\rangle_t\Big\} + o\Big(\sqrt{\mathrm{d}t}\Big) \tag{II.23}$$

$$\approx . \tag{II.24}$$

II.iii Linear quantum trajectories

Appendix A

Superposition of states VS statistical mixture

In order to illustrate the concepts of superposition of states and statistical mixture, let us consider to observables A, B such that

$$\forall i \in \{1,2\}, \quad A\left|\phi_i\right\rangle = a_i\left|\phi_i\right\rangle,$$
$$\forall i \in \{1,2\}, \quad B\left|\psi_i\right\rangle = b_i\left|\psi_i\right\rangle.$$

We can define a state $|\Psi\rangle$ in superposition of the eigenstates of the observable B:

$$\boxed{|\Psi\rangle = \lambda_1\left|\psi_1\right\rangle + \lambda_2\left|\psi_2\right\rangle}, \tag{A. 1}$$

where $(\lambda_1, \lambda_2) \in \mathbb{C}^2$ such that $|\lambda_1|^2 + |\lambda_2|^2 = 1$. We thus observe the result b_i with following probability

$$\forall i \in \{1,2\}, \quad \mathbb{P}_\Psi[b_i] = |\langle\psi_i|\Psi\rangle|^2 = |\lambda_i|^2 \equiv p_i. \tag{A. 2}$$

If now we measure the system in order to find one of the eigenvalues of A, a_1 for example, we find

$$\begin{aligned}\mathbb{P}_\Psi[a_1] &= |\langle\phi_1|\Psi\rangle|^2 \\ &= |\lambda_1\langle\phi_1|\psi_1\rangle + \lambda_2\langle\phi_1|\psi_2\rangle|^2.\end{aligned} \tag{A. 3}$$

Moreover, by observing that $\forall i \in \{1,2\}, \quad \mathbb{P}_{\psi_i}[a_1] = |\langle\phi_1|\psi_i\rangle|^2$, we finally find that

$$\boxed{\mathbb{P}_\Psi[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1] + 2\,\mathrm{Re}\{\lambda_1\lambda_2^*\langle\phi_1|\psi_1\rangle\langle\phi_1|\psi_2\rangle^*\}}. \tag{A. 4}$$

We will now prepare a state with the same probabilities $\mathbb{P}[b_i] \equiv p_i$ but which is completely different.

A statistical mixture can be well understood by the following thought experiment:

- We want to prepare N states in total using the eigenstates $\{\psi_1, \psi_2\}$ of B.
- We prepare n_1 states in $|\psi_1\rangle$.
- We prepare n_2 states in $|\psi_2\rangle$, so $n_1 + n_2 = N$.

At this point, we have a set of N systems that we can use. However, if we pick a state randomly the probability to pick $|\psi_i\rangle$ depends on the fraction of it $p_i = \frac{n_i}{N}$. This fraction p_i coincides

theoretically with the previous probability $P_\Psi[b_i]$ for the superposition of states: it represents the probability to find the state $|\psi_i\rangle$ from the general state. If we now use the observable A to measure the system we picked, the probability to find the result a_1 becomes

$$\mathbb{P}_{\text{sm}}[a_1] = \frac{n_1\mathbb{P}_{\psi_1}[a_1] + n_2\mathbb{P}_{\psi_2}[a_1]}{N} = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1],$$

by using the law of total probability. Therefore,

$$\boxed{\mathbb{P}_{\text{sm}}[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1]}. \tag{A. 5}$$

By comparing the last two boxed equations, one can realize that the mathematical expression for the same probability is written differently depending on if we use a superposition of states or a statistical mixture. However, the probability of the statistical mixture is present in the expression of probability of the superposition of states. This difference of formula for the same probability therefore proves that the Dirac formalism $|\psi\rangle$ cannot represent the kind of system whose probability comes from a statistical mixture. It is in fact impossible to represent a statistical mixture with a pure state. Instead, we need to use the density matrix

$$\rho_{\text{sm}} = p_1 \left|\psi_1\right\rangle\!\left\langle\psi_1\right| + p_2 \left|\psi_2\right\rangle\!\left\langle\psi_2\right|, \tag{A. 6}$$

so that,

$$\mathbb{P}_{\rho_{\text{sm}}}[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1], \tag{A. 7}$$

because by definition $\mathbb{P}_{\rho_{\text{sm}}}[a_1] = \text{Tr}\left[\rho_{\text{sm}} \left|\phi_1\right\rangle\!\left\langle\phi_1\right|\right]$.

Appendix B

Representations of quantum mechanics

Brief complement of quantum mechanics

An average is defined in quantum mechanics by

$$\begin{aligned}\langle O\rangle_{\psi} &= \langle\psi|O|\psi\rangle\,,\\ \langle O\rangle_{\rho} &= \mathrm{Tr}\left[\rho O\right],\end{aligned}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

B.i Schrödinger and Heisenberg pictures

In the Schrödinger picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$, where $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. Nevertheless, the following results can be derived in the general case of a time dependent Hamiltonian, [3]. Considering the average value of an observable $A_S(t)$ in the Schrödinger picture (in this picture, the time-dependence of an observable can only be explicit), we find

$$\begin{aligned}\langle A(t)\rangle &= \langle\psi(t)|A_S(t)|\psi(t)\rangle\\ &= \langle\psi(t_0)|U^{\dagger}(t,t_0)A_S(t)U(t,t_0)|\psi(t_0)\rangle &\text{(B. 1)}\\ &= \langle\psi|A_H(t)|\psi\rangle_H\,. &\text{(B. 2)}\end{aligned}$$

This calculation naturally reveals the Heisenberg representation of observables and states. Indeed, we have that $|\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time t_0. Furthermore, we can link both representation at all time by calculating the evolution of an observable in the Heisenberg picture:

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}}. \tag{B. 3}$$

This equation leads to the famous *Ehrenfest theorem* analogous in quantum mechanics:

$$\boxed{\frac{\mathrm{d}\langle A\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\langle[H(t), A(t)]\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle}. \tag{B. 4}$$

We observe that if the observable doesn't depend explicitly on time, then the dynamics of the operator is fully described in the Heisenberg picture.

B.ii Interaction picture

As mentioned in the Section I.iii, we can represent the coupling between S and E with an interaction Hamiltonian such that

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{B. 1}$$

Knowing that $|\psi(t)\rangle_S = U(t,t_0)\,|\psi\rangle_S$, we can write that

$$\rho_S(t) = U(t,t_0)\rho_S(t_0)U^\dagger(t,t_0). \tag{B. 2}$$

The evolution operator U is the solution of the Schrödinger equation for the total system T. We know from Section I.i.1 that we can write the solution of the Schrödinger equation when having a time-independent Hamiltonian as an exponential. However here, the total Hamiltonian H_T is obviously not time independent (only H_0 is assumed constant in time); nevertheless, a unitary solution still exists in this case [3]. We know that the free system (dynamically governed by H_0) follows unitary dynamics, which is described by $U_0(t,t_0) = e^{-\frac{i}{\hbar}H_0(t-t_0)}$. Thus, we can define the interaction unitary operator U_I as

$$U_I(t,t_0) = U_0^\dagger(t,t_0)U(t,t_0) \tag{B. 3}$$

$$\Leftrightarrow U(t,t_0) = U_0(t,t_0)U_I(t,t_0) \tag{B. 4}$$

If we now try to calculate the average of an operator (in the total space $\mathcal{H}_T$) using the Schrödinger picture, we can find a new interesting relation by injecting the Eq. (B. 4) in the Eq. (B. 2) and then into the following average:

$$\begin{aligned}\langle O(t)\rangle &= \mathrm{Tr}\left[\rho_S(t)O_S(t)\right] \\ &= \mathrm{Tr}\left[\underbrace{U_I(t,t_0)\rho_S(t_0)U_I^\dagger(t,t_0)}_{\tilde{\rho}(t)}\ \underbrace{U_0^\dagger(t,t_0)O_S(t)U_0(t,t_0)}_{\tilde{O}(t)}\right].\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\rho}(t) = U_I(t,t_0)\rho_S(t_0)U_I^\dagger(t,t_0), \tag{B. 5}$$

$$\tilde{O}(t) = U_0^\dagger(t,t_0)O_S(t)U_0(t,t_0). \tag{B. 6}$$

We can now reverse the Eq. (B. 2) and then use the Eq. (B. 4) to get

$$\rho_S(t_0) = U^\dagger(t,t_0)\rho_S(t)U(t,t_0) = U_I^\dagger(t,t_0)U_0^\dagger(t,t_0)\rho_S(t)U_0(t,t_0)U_I(t,t_0),$$

and, by injecting this into the Eq. (B. 5), we get

$$\tilde{\rho}(t) = U_0^\dagger(t,t_0)\rho_S(t)U_0(t,t_0). \tag{B. 7}$$

This result is obvious because a density matrix is also an operator and thus respects the Eq. (B. 6). In the end, we find the results used in the main part of this document:

$$\boxed{\tilde{\rho}(t) = U_I(t,t_0)\rho_S(t_0)U_I^\dagger(t,t_0) = U_0^\dagger(t,t_0)\rho_S(t)U_0(t,t_0)}. \tag{B. 8}$$

By working on the Liouville-von Neumann equation, one can show that by jumping into the interaction picture the dynamics do not depend anymore on the free Hamiltonian H_0:

$$\left(\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\frac{i}{\hbar}[H_T,\rho]\right) \Leftrightarrow \left(\frac{\mathrm{d}\tilde{\rho}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I,\tilde{\rho}\right]\right), \tag{B. 9}$$

where $\tilde{H}_I = U_0^\dagger(t,t_0)H_I U_0(t,t_0)$ as defined. It is thus usual to work in this representation to get rid of free dynamics terms, and then jump back to the Schrödinger picture to recover the proper dynamics of the system.

Further discussion

We used in Eq. (I.39) the following approximation: $U_I(t+\mathrm{d}t,t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t}$, and we would like to explain it here. As seen in Eq. (B. 5), a state evolves in the interaction picture with a relation depending on U_I, which has been defined as $U_I = U_0^\dagger U$. It is possible to show from this definition that U_I is subject to the following differential equation

$$\frac{\mathrm{d}U_I}{\mathrm{d}t} = -\frac{i}{\hbar}\tilde{H}_I U_I, \tag{B. 10}$$

where we recall that $\tilde{H}_I = U_0^\dagger H_I U_0$ by definition of the interaction picture. This equation is subject to the condition $U_I(t_0,t_0) = \mathbb{I}$. We can now work on the operator $U_I(t+\mathrm{d}t,t_0)$, which can be expressed by a Taylor expansion up to order $\mathrm{d}t$ as

$$U_I(t+\mathrm{d}t,t_0) = U_I(t,t_0) + \frac{\mathrm{d}U_I}{\mathrm{d}t}(t,t_0)\mathrm{d}t + o(\mathrm{d}t). \tag{B. 11}$$

We can then inject Eq. (B. 10) into this expansion to get

$$U_I(t+\mathrm{d}t,t_0) = U_I(t,t_0) - \frac{i}{\hbar}\tilde{H}_I(t,t_0)U_I(t,t_0)\mathrm{d}t + o(\mathrm{d}t). \tag{B. 12}$$

By setting $t = t_0$ and using the condition on unitary operator $U_I(t_0,t_0)$ we obtain

$$U_I(t+\mathrm{d}t,t) = \mathbb{I} - \frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t + o(\mathrm{d}t). \tag{B. 13}$$

Note that technically, $\tilde{H}_I(t_0,t_0) = H_I(t_0)$ for the same reason because of U_0. However we are interested into using the expression of $\tilde{H}_I$ in the calculation of Section I.iii and so, we decide to keep the expression in the interaction picture. We can observe that this expansion corresponds to the expansion of the exponential at the same order, indeed

$$e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t} = \mathbb{I} - \frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t + o(\mathrm{d}t), \tag{B. 14}$$

thus proving the expression we used:

$$U_I(t+\mathrm{d}t,t) \approx e^{-\frac{i}{\hbar}\tilde{H}_I(t)\mathrm{d}t}. \tag{B. 15}$$

Appendix C

A taste of quantum stochastic calculus and input-output formalism

blablabla

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